INSTITUTO TECNOLÓGICO DE CD. GUZMÁN

POWER CONDITIONING SYSTEMS FOR RENEWABLE ENERGY SOURCES

ACTIVITY:

MATHEMATICAL MODEL FOR PV CELL

STUDENT:

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INTRUCTION

In this work, MATLAB code was developed based on the mathematical model of a photovoltaic cell proposed by Saleem et al. (2020). The objective is to analyze the electrical behavior of a solar panel by generating current-voltage (I-V) and power-voltage (P-V) curves, evaluating the effects of solar irradiance, temperature, and shunt resistance on its performance.

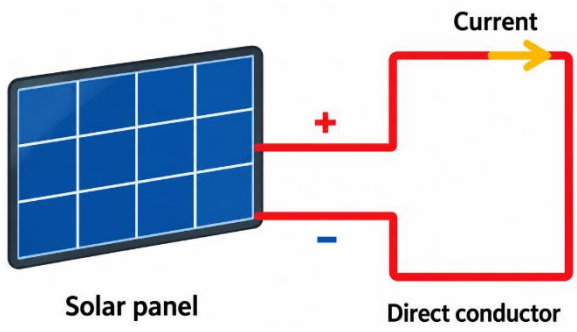
SOME BASIC CONCEPTS FOR UNDERSTANDING THE MATHEMATICAL MODEL

Solar Irradiance

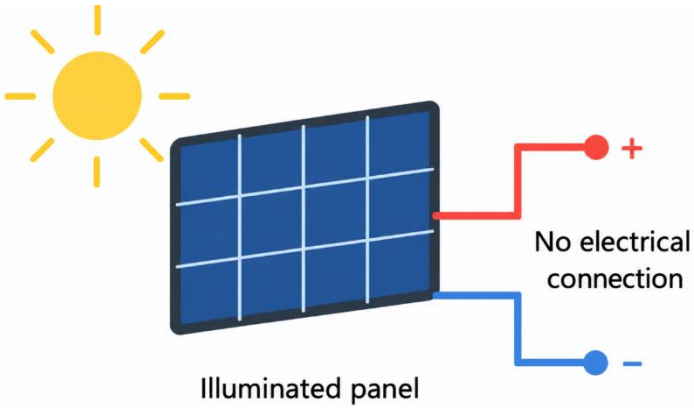
Solar irradiance is the amount of solar power received per unit area of a surface. It is represented by the symbol G and measured in watts per square meter (W/m²). In photovoltaic systems, higher solar irradiance generally increases the electrical current generated by the solar panel, resulting in greater power output.

As solar irradiance increases, the solar panel can deliver more electrical current because it receives more solar radiation.

Short-circuit current (I_{sc}) It is the current produced by a solar panel when the output voltage is zero $V = 0$.



Open-circuit voltaje (V_{oc}) It is the maximum voltage produced by a solar panel when no electrical load is connected and no current flows.



SOLAR PANEL CURRENT

$$I = I_{ph} - I_d - I_{sh}$$

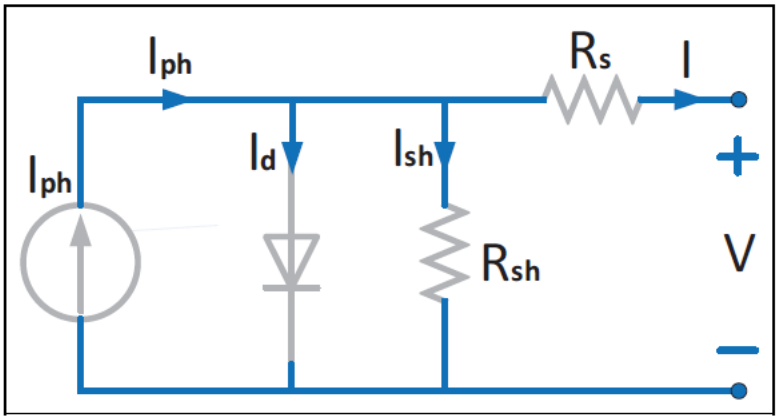


Figure 2. Real equivalent circuit for photovoltaic cell.

The current delivered by a solar panel is the photocurrent (I_{ph}) generated by sunlight, minus two internal currents: the diode current (I_d) and the shunt current (I_{sh}).

When photons reach the semiconductor and are absorbed, they can generate electron-hole pairs. The internal electric field of the solar cell helps separate these charges.

Generally, the higher the solar irradiance received by the panel, the greater the photocurrent generated.

The article uses the following equation:

$$I_{ph} = [I_{sc} + K_i(T - 298)]\left(\frac{G}{1000}\right)$$

Where:

Symbol	Meaning
I_{ph}	Generated photocurrent (A).
I_{sc}	Reference short-circuit current (A).
K_i	Current temperature coefficient (A/°C).
T	Cell operating temperature (K).
298	Reference temperature, approximately 25 °C (298 K).
G	Solar irradiance (W/m^2).
1000	Reference solar irradiance (1000 W/m^2).
I_{sh}	Shunt current, also called leakage current (A).
R_{sh}	Represents the current leakage paths inside a photovoltaic cell.

MATLAB CODE FOR PHOTOVOLTAIC MODEL

MATLAB Code

```
%% PHOTOVOLTAIC MODEL - FIGURES 11 TO 16
clear; clc; close all;

% Solar panel parameters
Isc=5.85; Voc=39.80; Rs=0.221; Rsh=415.405;
Ki=0.0032; Tn=298; n=1.3; Ego=1.1; Ns=54;
k=1.38e-23; q=1.6e-19; V=linspace(0,45,600);

% Conditions, colors and styles
values={[400 600 800 1000],[15 25 45 70],[415.405 100 10 5]};
palette=[0.00 0.20 0.45;
         0.00 0.35 0.65;
         0.25 0.55 0.80;
         0.60 0.78 0.92];
styles={'-','--',':','-.'};
names={'irradiance','temperature','shunt resistance'};
units={'W/m^2','°C','\Omega'};
folder='PV_Figures_PDF';
if ~exist(folder,'dir'), mkdir(folder); end

for a=1:3
    I=zeros(4,numel(V));
    for j=1:4
        G=1000; T=298; R=Rsh;
        if a==1, G=values{a}(j); end
        if a==2, T=values{a}(j)+273; end
        if a==3, R=values{a}(j); end

        Iph=(Isc+Ki*(T-298))*G/1000;
        Irs=Isc/expm1(q*Voc/(n*k*Tn*Ns));
        Io=Irs*(T/Tn)*3*exp(q*Ego/(n*k*)(1/Tn-1/T));
        if a==2
            VocT=Voc-.23*(T-298);
            Io=(Iph-VocT/R)/expm1(q*VocT/(n*k*T*Ns));
        end

        x=Iph;
        for m=1:numel(V)
            for it=1:100
                z=q*(V(m)+x*Rs)/(n*k*T*Ns);
                e=exp(min(z,700));
                f=Iph-Io*(e-1)-(V(m)+x*Rs)/R-x;
                df=-Io*e=q*Rs/(n*k*T*Ns)-Rs/R-1;
                xn=x-f/df;
                if abs(xn-x)<1e-10, x=xn; break; end
                x=xn;
            end
            I(j,m)=max(x,0);
        end
    end

    for b=1:2
        fig=9+2*a+b;
        h=figure(fig); clf(h); hold on; box on;
        set(h,'Color','w','Units','centimeters','Position',[3 3 17 11]);
        Y=I; tipo='I-V'; eje='Current (A)';
        if b==2, Y=I.*V; tipo='P-V'; eje='Power (W)'; end

        for j=1:4
            plot(V,Y(j,:),'Color',palette(j,:), ...
                'LineStyle',styles{j},'LineWidth',2);
        end

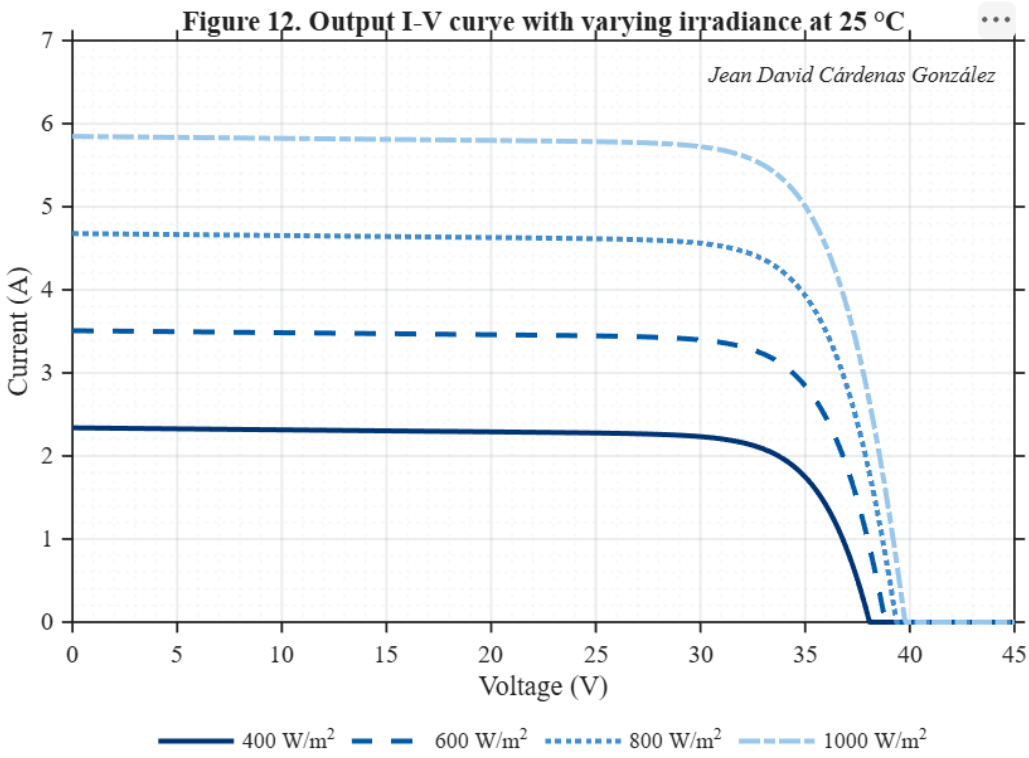
        labels=arrayfun(@(x)sprintf("%g %s",x,units{a}), ...
            values{a},'UniformOutput',false);
        cond={'at 25 °C','at 1000 W/m²','at 25 °C and 1000 W/m²'};

        xlabel('Voltage (V)'); ylabel(eje); xlim([0 45]);
        if b==1, ylim([0 7]); else, ylim([0 220]); end
        title(sprintf('Figure %d. Output %s curve with varying %s %s', ...
             fig,tipo,names{a},cond{a}),'FontSize',10);
        legend(labels,'Location','southwest','Box','off','FontSize',9);

        ax=gca; ax.FontName='Times New Roman'; ax.FontSize=10;
        ax.LineWidth=.9; ax.TickDir='out'; ax.XTick=0:5:45;
        if b==1, ax.YTick=0:1:7; else, ax.YTick=0:20:220; end
        grid on; grid minor;
        ax.GridColor=[.55 .58 .62]; ax.GridAlpha=.18;
        ax.MinorGridColor=[.65 .68 .72]; ax.MinorGridAlpha=.07;
        ax.MinorGridLineStyle=':';

        text(.98,.96,'Jean David Cárdenas González', ...
            'Units','normalized','HorizontalAlignment','right', ...
            'VerticalAlignment','top','FontAngle','italic', ...
```

GRAPHS PHOTOVOLTAIC MODEL



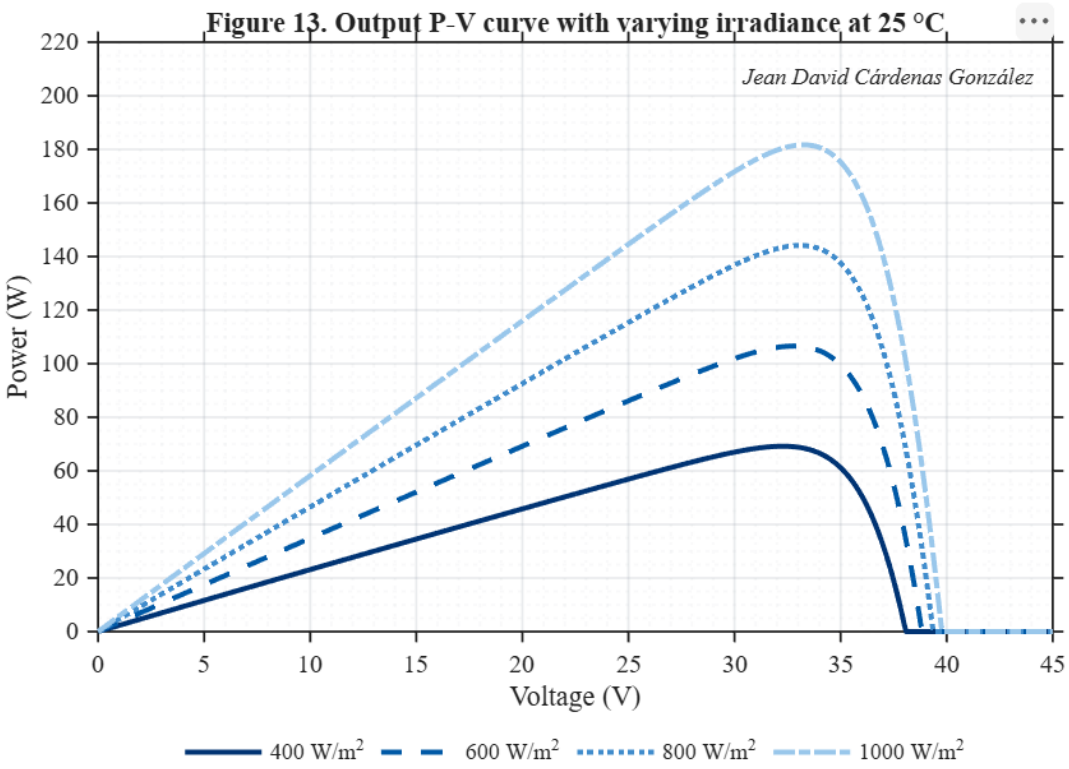


Figure 14. Output I-V curve with varying temperature at 1000 W/m²

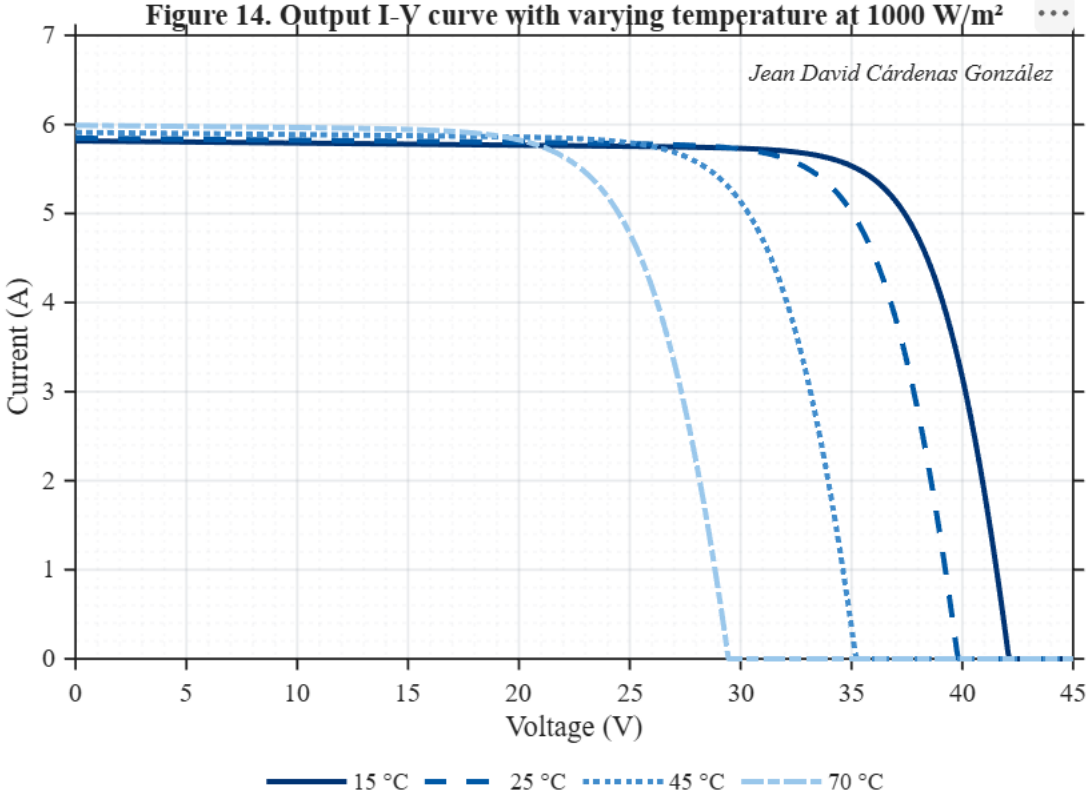


Figure 15. Output P-V curve with varying temperature at 1000 W/m²

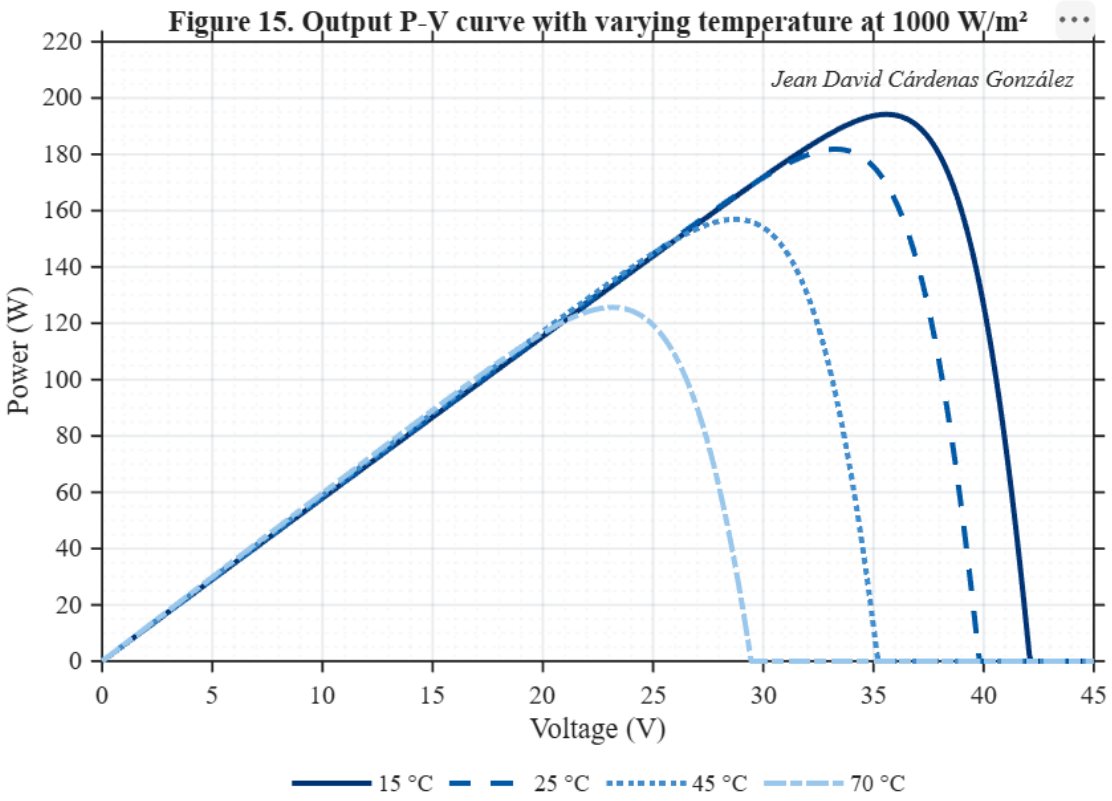


Figure 16. Output I-V curve with varying shunt resistance at 25 °C and 1000 W/m²

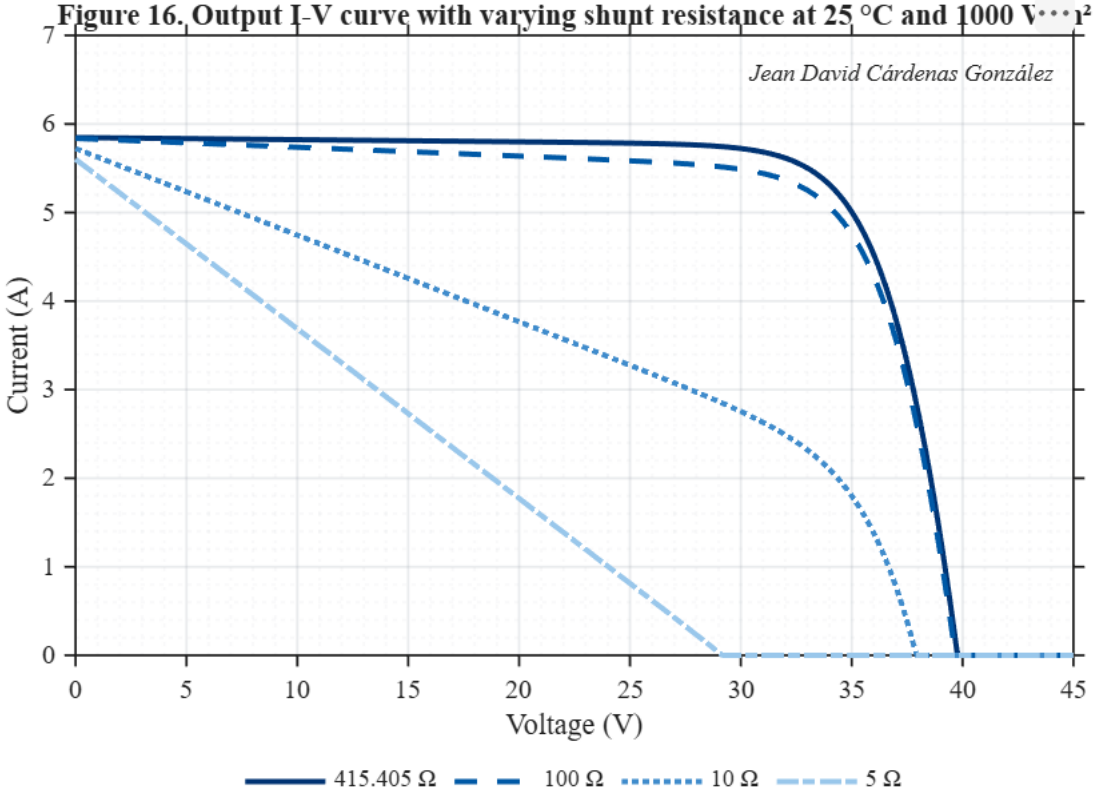
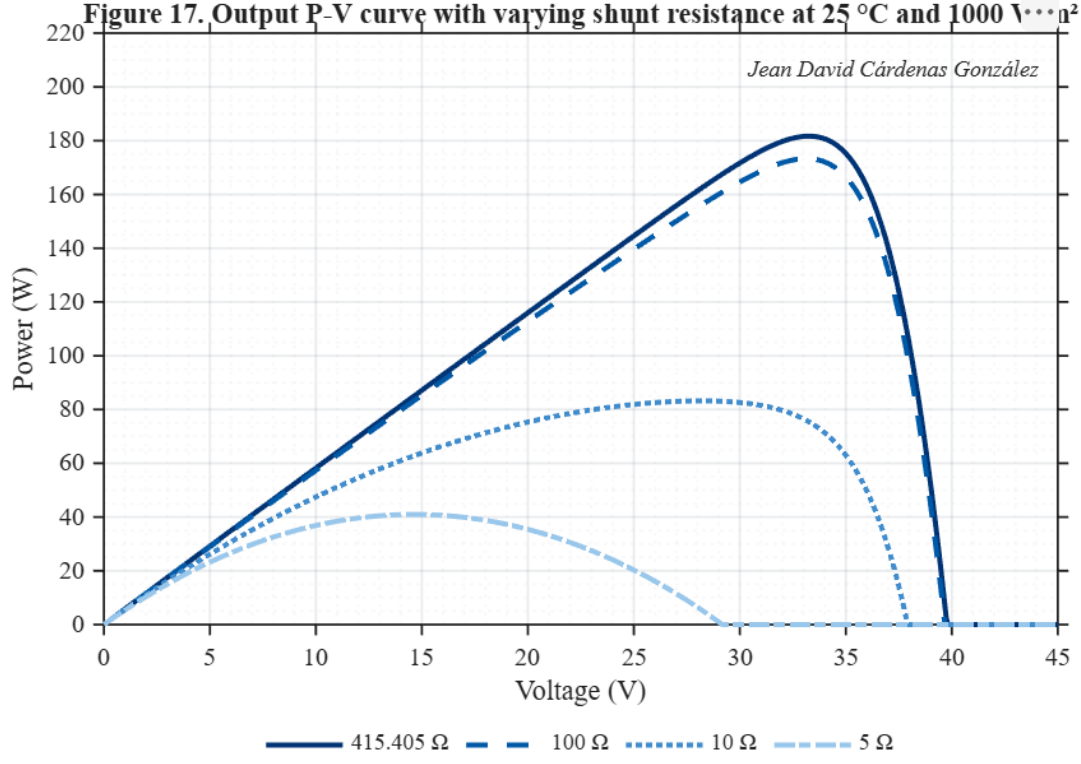


Figure 17. Output P-V curve with varying shunt resistance at 25 °C and 1000 W/m²



Conclusion

The MATLAB simulation allowed us to analyze the electrical behavior of a photovoltaic panel under different operating conditions. The results showed that increasing solar irradiance increases the current and power output, while higher temperatures reduce the voltage and maximum power. Additionally, decreasing the shunt resistance increases internal losses and reduces the panel's performance. Overall, the mathematical model provides a useful tool for understanding the factors that influence photovoltaic energy generation.